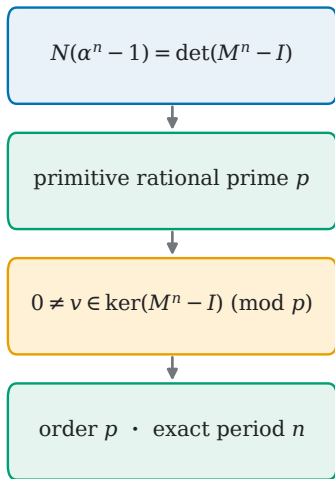
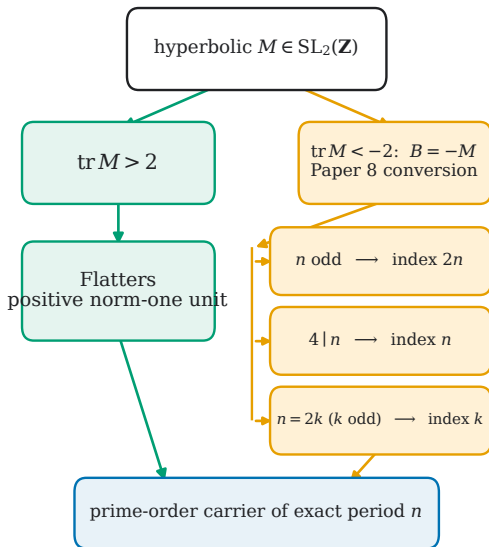


a Arithmetic bridge



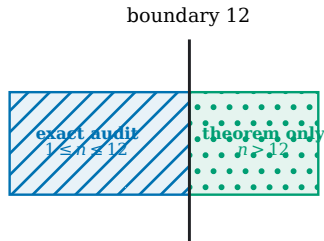
if $r = \dim \ker(M^n - I)$: $(p^r - 1)/n$ cycles

b Sign and parity routing



c Evidence boundary

manifest PASS



computed tail periods: 0
imported primitive-divisor theorem
+ separate parity proof